COLLEGE COMMUNITY ENGAGEMENT PROJECT (CEP) REPORT

RuralRise / GramTech

Technology Support for Rural Startups — Project Dossier

Topic:	Technology Support for Rural Startups	**Date:**	Academic Year 2025-2026
Application:	RuralRise / GramTech Web Application	**Tech Stack:**	Standard Java 21 + HTML5/CSS3/JS
Architecture:	com.sun.net.httpserver.HttpServer (Zero External JARs)	**Execution URL:**	http://localhost:8080

Project Purpose & Synopsis:
RuralRise / GramTech is an educational web prototype developed to empower village entrepreneurs, women-led Self-Help Groups (SHGs), traditional craftspeople, and smallholder farmers. The system removes middleman barriers by training founders in digital payments, WhatsApp commerce, practical costing guidance, and actionable enterprise blueprints. Designed for academic evaluation, the software runs locally on a built-in Java HTTP server with zero database setup and zero internet dependency.

1. Problem Statement & CEP Objectives

Rural enterprises produce high-quality agro-products, handloom textiles, and handicrafts, yet experience high failure rates and marginal revenues. Our Community Engagement Project identified key structural deficits:

- **Middlemen Dependency:** Village producers lose 40% to 65% of market value to intermediaries because they lack direct digital buyer access.
- **Fear of Digital Fintech:** Apprehension regarding UPI payments, online transactions, and lack of verified safety awareness.
- **Absent Financial Discipline:** Failure to separate home grocery expenses from shop capital, causing working capital exhaustion.

Core Objectives of the RuralRise Software:

1. Deliver 11 structured, non-technical learning modules on pricing, digital sales, and cybersecurity.
2. Provide 8 turnkey rural enterprise blueprints with specific capital, skill, and technology requirements.
3. Present a modern, accessible web portal with instant search, category filtering, and dual view modes.
4. Enable 100% offline demonstration reliability powered by a standard Java HTTP server.

2. Technical Architecture & Java Implementation

To guarantee 100% offline demonstration reliability during college evaluations without requiring heavy installations (like Spring Boot or MySQL), the application uses standard Java 21 built-in libraries:

Component	Source File	Role & Technical Implementation

HTTP Web Server	GramTechServer.java	Utilizes standard com.sun.net.httpserver.HttpServer. Handles automatic port fallback (8080/8081), CORS headers, and routes static assets and the /api/data endpoint.
Data Repository	logic/GramTechData.java	Static in-memory data store containing all 11 guides, 8 business models, 5 digital tool pillars, and FAQs, outputting JSON via /api/data.
Interactive Frontend	web/index.html web/styles.css web/app.js	Responsive Material 3 user interface with instant keyword search, category filter pills, modal drawers, and Dual View switcher (Desktop Portal vs Phone Frame).
1-Click Runner	run_web_app.bat	Automated batch file that detects JDK, compiles source files via javac, starts the server, and launches the browser.

3. Detailed Walkthrough of Application Screens & Sections

Screen 1: ■ Home Screen & Dual View Switcher

Purpose: Serves as the landing hub providing high-level context, quick statistics, and direct action shortcuts.
• **Hero Section:** Clear mission banner and call-to-action buttons (*Explore Guides*, *Business Ideas*, *Digital Tools*).
• **Live Statistics Strip:** Displays 11+ Practical Guides, 8 Business Models, 5 Digital Tool Pillars, and 100% Java Backend.
• **Dual View Switcher:** Located in the top bar. Enables college evaluators to toggle between **Desktop Portal Mode** (multi-column dashboard for projector screens) and **Smartphone Mockup Frame Mode** (420px mobile mockup with speaker notch and bottom navigation).

Screen 2: ■ Learning Resources (Startup Guidance)

Contains **11 practical guides** designed with simple language and actionable steps. Includes real-time keyword search and 4 category filter tabs (*All Topics*, *Business Basics*, *Digital Skills*, *Financial Awareness*):

Guide Title	Category	Key Takeaway & Practical Advice
How to Start a Small Business	Business Basics	Identifying village needs, choosing proprietorship vs SHG, and obtaining free Udyam MSME certification.
Basic Business Planning	Business Basics	Writing a simple 1-page business plan covering capital, operating expenses, and monthly break-even targets.
Understanding Customers	Business Basics	Interviewing 20 local buyers at weekly haats before purchasing inventory to validate actual product demand.
Pricing Your Products	Business Basics	True Price = (Raw Materials + Packaging + Fuel + Fair Labor Wages + 25% Profit Margin). Never treat own labor as free.
Managing Business Expenses	Business Basics	Separating household cash from shop funds to prevent working capital depletion.
Digital Marketing Basics	Digital Skills	Taking crisp daylight product photos with smartphones and posting in town WhatsApp groups.
Social Media & WhatsApp	Digital Skills	Using WhatsApp Business catalogs, automated greetings, and order management tags.
Online Payments & UPI Safety	Digital Skills	UPI Golden Rule: Entering PIN is strictly for SENDING money, never for receiving. Using soundboxes for audio alerts.
Google Business & Visibility	Digital Skills	Creating free Google Maps listings with phone numbers to attract highway travelers and wholesale buyers.
Understanding Startup Costs	Finance	Differentiating one-time fixed capital (tools) from 3 months of rolling working capital (seeds/materials).
Loan Awareness & Credit Health	Finance	Building verifiable digital transaction statements via UPI deposits to qualify for low-interest formal credit.

Screen 3: ■ Rural Business Ideas (Blueprints)

Features **8 turnkey rural business models** showing required capital, skills, and tools:

Business Model	Capital Tier	Technology Stack & Revenue Model
Organic Farming & Direct Sales	Low-Medium	WhatsApp buyer groups, UPI advance QR payments; delivers weekly vegetable baskets directly to city apartment complexes.
Food Processing & Pickles	Medium	Heat sealers, FSSAI digital registration, Instagram Reels; converts perishable mangoes/amla into high-margin packaged jars.
Handicrafts & Artisanal Decor	Low	Smartphone photography, ONDC e-commerce catalogs, India Post tracking; elevates bamboo/pottery into national gifting items.
Smart Dairy Management	Medium-High	Digital fat/SNF milk testers, cattle health apps, SMS ledger; fair price realization paid directly into bank accounts.

Rural Tourism & Farmstays	Medium	Google Maps listing, online advance booking; hosts urban families for authentic village cuisine and farm culture.
Agri Equipment Rental	High	GPS tracking, mobile booking register; Custom Hiring Centre renting mini tractors and power sprayers to smallholders.
Rural Last-Mile Delivery	Low	Delivery tracking apps, digital signatures; acts as trusted village cluster parcel pickup and dispatch hub.
Digital Village Service Center	Low-Medium	Desktop PC, biometric fingerprint scanner, broadband; CSC providing Aadhaar, utility bills, and scholarship filings.

Screen 4: ■■ Essential Digital Tools Directory

Structured into **5 practical functional pillars** enabling rural startups to operate digitally without IT overhead:

- **1. Communication (■):** WhatsApp Business, Voice/Video Calls, Email → Free in-app catalogs and automated welcome messages.
- **2. Marketing (■):** Instagram Reels, Facebook Community Groups, Canva Graphic Creator → Zero-cost viral video promotion.
- **3. Payments (■):** UPI QR Code Standees, BHIM / PhonePe / GPay, UPI Soundbox → Instant audio settlement alerts and zero bad debts.
- **4. Productivity (■):** Google Sheets, Digital Ledger (Khata App), Cloud Drive → Automated sales tracking and paperless debt books.
- **5. Learning (■):** YouTube Video Tutorials, Kisan Suvidha App, Skill India Portal → Free crop diagnosis and machine maintenance tutorials.

Screen 5: ■■ About & Evaluator FAQs

Documents academic objectives, Community Engagement Project problem statement, and interactive evaluator FAQ accordion addressing common viva questions.

4. Backend REST API Endpoint Specification

The native Java HTTP server exposes a clean JSON endpoint with full CORS headers:

Endpoint	HTTP Method	Parameters / Body	Description & Output
/api/data	GET	None	Full JSON catalog containing all 11 guides, 8 business models, 5 digital tool pillars, and FAQs.
/	GET	Path (HTML/CSS/JS)	Serves static web application files (index.html, styles.css, app.js) with correct MIME types.

5. College Presentation & Viva Cheatsheet

Prepared answers for common questions asked by academic examiners and evaluation committees:

Frequently Asked Question	Recommended Technical & Academic Response
What inspired this topic?	Rural entrepreneurs frequently produce premium organic and handcrafted goods but suffer from low margins due to intermediaries. Our CEP software demonstrates how affordable smartphones, UPI, and online catalogs bridge rural enterprises directly to high-value markets.
Why use standard Java instead of Spring Boot or Node.js?	College demonstrations require zero installation friction. Using standard Java's built-in com.sun.net.httpserver eliminates heavy build tools, 500MB dependency downloads, and setup failures. It starts in under 1 second on any machine with JDK.
Where is the database located?	To satisfy academic offline evaluation constraints, all educational content is structured as an in-memory JSON data repository in GramTechData.java. This avoids database crash risks and makes the prototype 100% portable.
What is the purpose of Dual View Mode?	It allows evaluators to assess both desktop portal ergonomics (for wide laptop/projector presentations) and responsive mobile UX (simulating how a village artisan views the application on a smartphone).

Project Submission Confirmation:
Project: Technology Support for Rural Startups (RuralRise / GramTech)
Status: Fully Implemented, Compiled & Verified • Active at http://localhost:8080
Ready for Academic Evaluation & Presentation.